

2.2 The Inverse of a Matrix

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C such that

$$CA = I \text{ and } AC = I$$

where $I = I_n$, the $n \times n$ identity matrix. In this case, C is an inverse of A .
 In fact, C is uniquely determined by A , because if B were another inverse of A , then $B = BI$ (Theorem 2(e)) $= B(AC)$ (Given) $= (BA)C$ (Theorem 2(a)) $= IC$ (By defn.) $= C$ (Theorem 2(e)).

This unique inverse is denoted by A^{-1} , so that $A^{-1}A = I$ and $AA^{-1} = I$.

A matrix that is not invertible is sometimes called a singular matrix, and an invertible matrix is called a nonsingular matrix.

Example 1: If $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$, then

$$AC = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = [AC_1 \quad AC_2] \quad (\text{By defn. of matrix multiplication})$$

$$AC_1 = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot (-7) + 5 \cdot 3 \\ (-3) \cdot (-7) + (-7) \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(Row-Vector Rule)

$$AC_2 = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -5 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot (-5) + 5 \cdot 2 \\ (-3) \cdot (-5) + (-7) \cdot 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(Row-Vector Rule for Computing Ax)

$$\therefore AC = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$CA = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = [CA_1 \quad CA_2] \quad (\text{By defn. of matrix multiplication})$$

$$CA_1 = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix} = 2 \begin{bmatrix} -7 \\ 3 \end{bmatrix} + (-3) \begin{bmatrix} -5 \\ 2 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} -14 \\ 6 \end{bmatrix} + \begin{bmatrix} 15 \\ -6 \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (\text{Vector addition})$$